(KTXFI2EBNF) Physics II. Homework No. 5 Solutions

December 19, 2025

1. What forms of radioactivity exist? What is neutron radiation? How can the environment be protected from different forms of radiation?

   The main forms of radioactivity are alpha (α), beta (β), and gamma (γ) radiation. Alp radiation consists of helium nuclei and has low penetration ability but strong ionizing power. Beta radiation is composed of high-energy electrons or positrons and has moderat penetration. Gamma radiation is electromagnetic radiation with very high penetration ability.

   Neutron radiation consists of free neutrons, which are uncharged particles. Due to the lack of electric charge, neutrons can penetrate deeply into matter and cause secondary radiation through nuclear reactions, making them particularly dangerous.

   Environmental protection against radiation includes shielding with suitable materials (paper for alpha, aluminum for beta, lead or concrete for gamma radiation), limiting exposure time, increasing distance from sources, and strict regulation and monitoring of radioactive materials.

2. What nuclear reactors do you know? What advanced reactors are currently under development? What type of nuclear reactor should Hungary's third nuclear power plant have?

   Common nuclear reactor types include pressurized water reactors (PWR), boiling water reactors (BWR), heavy water reactors, gas-cooled reactors, and fast neutron reactors. Most currently operating nuclear power plants worldwide use PWR technology due to its reliability and safety features.

   Advanced reactors under development include Generation IV reactors such as fast breeder reactors, molten salt reactors, high-temperature gas-cooled reactors, and small modular reactors (SMRs). These designs aim to improve safety, efficiency, fuel utilization, and waste reduction.

   Hungary's third nuclear power plant should likely use a modern pressurized water reactor or a small modular reactor, as these technologies are well-established, meet internation safety standards, and integrate well with Hungary's existing nuclear infrastructure.

3. What do you know about the creation of gold, platinum, uranium, and plutonium? Where are these elements created?

   Gold and platinum are primarily formed during extreme astrophysical events such as super explosions and neutron star mergers, where high neutron flux enables rapid neutron captu processes. Uranium is also created through similar stellar processes, mainly via slow and rapid neutron capture in massive stars before and during supernova explosions. Plut does not occur in significant natural quantities on Earth and is mainly produced artific in nuclear reactors through neutron capture by uranium isotopes.

4. Is it possible to build nuclear-powered cars or spacecraft? What types of nuclear reactors are used in nuclear submarines and aircraft carriers? What do you think about nuclear rockets, such as those depicted in *2001: A Space Odyssey*?

   Nuclear-powered cars are impractical due to safety, size, shielding, and regulatory chal Nuclear-powered spacecraft, however, are feasible and already exist in the form of radic thermoelectric generators and proposed nuclear thermal propulsion systems. Nuclear subm and aircraft carriers typically use compact pressurized water reactors designed for long operation without refueling. Nuclear rockets, such as those depicted in *2001: A Space*

Odyssey, are theoretically feasible and could provide high efficiency for deep-space missions. While technologically challenging, they represent a promising option for futu interplanetary travel.

5. What are quarks? What is the Higgs boson? What is the role and activity of CERN in Switzerland?
   Quarks are fundamental particles that form hadrons, such as protons and neutrons. They come in six types, called flavors, and interact through the strong nuclear force.
   The Higgs boson is a fundamental particle associated with the Higgs field, which gives mass to elementary particles through the Higgs mechanism.
   CERN is the European Organization for Nuclear Research, located in Switzerland. Its main role is to conduct high-energy particle physics research, operate large particle accelerators such as the Large Hadron Collider, and explore the fundamental structure of matter and the universe.

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